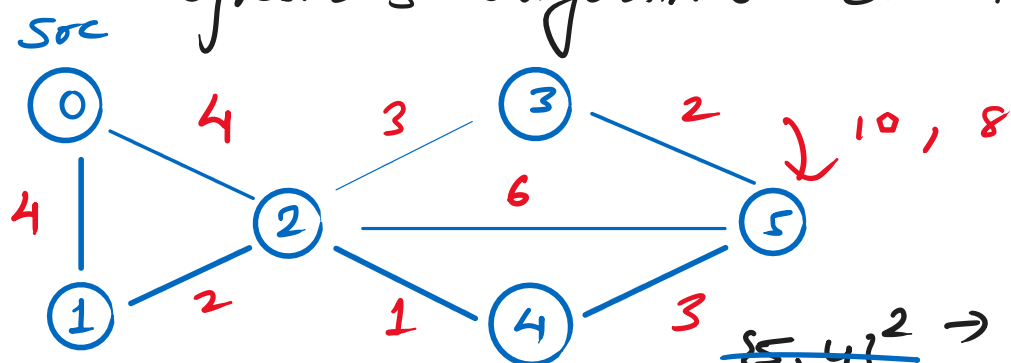


Dijkstra's Algorithm : Shortest Path (Set - DS)



Pq → erase x
set → erase

	4	4	7	5	8
0	✓	✓	✓	✓	✓
0	1	2	3	4	5

distance array

{5, 4}
{7, 3}
{8, 5}
{0, 0}
{1, 1}
{4, 2}
{10, 5}

Adj-List
 → 0 → {1, 4} {2, 4}
 1 → {0, 4} {2, 2}
 2 → {0, 4} {1, 2} {3, 3} {4, 1} {5, 6}
 3 → {2, 3} {5, 2}
 4 → {2, 1} {5, 3}
 5 → {2, 6} {3, 2}, {4, 3}

set → {dist, node}

erase() ∈ Log V
 removed, because we found a better path.

In the Dijkstra's algo using set, if we find that the adjacent node was already visited but with a greater distance, we 'erase' the earlier path value from the set and push the new value {d, n}.

This is done to reduce the no. of iterations.
 Or, else this earlier value will be again iterated uselessly.

Set → Ascending Order